RIG-SMOOTHNESS KILLS THE H^{-1} CONNECTING TERM OF THE NAIVE COTANGENT SEQUENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof reduces to injectivity at primes avoiding an ideal of definition, translates it through the presentations into the intersection identity J_q cap K^2_q = (KJ)_q, and derives it from rig-smooth implies rig-flat plus the H_1-regular conormal sequence lemma

Editorial note. The label lemma-exact-sequence-NL-rig-smooth names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the exact sequence of completed naive cotangent complexes for A -> B -> C and the induced map on H^{-1} (lemma-exact-sequence-NL of the same chapter, including its presentation diagram)
- presentations of adic topologically finite type algebras B = A{x_1,...,x_r}/J, C = B{y_1,...,y_s}/J' and C = A{x,y}/K over the completed polynomial rings
- rig-smooth implies rig-flat, so the ideal (K/J)_q is generated by a regular sequence at the relevant primes (lemma-rig-smooth-rig-flat of the same chapter)
- the conormal sequence of an ideal generated by a regular sequence is H_1-regular (More on Algebra, Lemma more-algebra-lemma-conormal-sequence-H1-regular-ideal and its section on ideals)

Source reference. Original reference 00S2: sources/stacks-restricted/pinned-original-source.tex, source lines 725–725. Prerequisite pointer invoked at proof line(s) 4568; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GH1: sources/stacks-restricted/pinned-original-source.tex, source lines 4460–4460. Prerequisite pointer invoked at proof line(s) 4594; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07CX: sources/stacks-chow/tags.txt, source lines 9392–9392. Prerequisite pointer invoked at proof line(s) 4599; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07D4: sources/stacks-chow/tags.txt, source lines 9399–9399. Prerequisite pointer invoked at proof line(s) 4589; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 07CU: sources/stacks-chow/tags.txt, source lines 9389–9389. Prerequisite pointer invoked at proof line(s) 4602; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $A \to B \to C$ be arrows in $WAdm^{Noeth}$ which are adic and topologically of finite type. If $B \to C$ is rig-smooth, then the kernel of the map*

$$H^{-1}(NL^\wedge_{B/A} \otimes_B C) \to H^{-1}(NL^\wedge_{C/A})$$

(see Lemma 00S2) is annihilated by an ideal of definition.

Proof. Let $\overline{\mathfrak{q}} \subset C$ be a prime ideal which does not contain an ideal of definition. Since the modules in question are finite it suffices to show that

$$H^{-1}(NL^\wedge_{B/A} \otimes_B C)_{\overline{\mathfrak{q}}} \to H^{-1}(NL^\wedge_{C/A})_{\overline{\mathfrak{q}}}$$

is injective. As in the proof of Lemma 00S2 choose presentations $B = A\{x_1, \ldots, x_r\}/J$, $C = B\{y_1, \ldots, y_s\}/J'$, and $C = A\{x_1, \ldots, x_r, y_1, \ldots, y_s\}/K$. Looking at the diagram in the proof of Lemma 00S2 we see that it suffices to show that $J/J^2 \otimes_B C \to K/K^2$ is injective after localization at the prime ideal $\mathfrak{q} \subset A\{x_1, \ldots, x_r, y_1, \ldots, y_s\}$ corresponding to $\overline{\mathfrak{q}}$. Please compare with More on Algebra, Lemma 07D4 and its proof. This is the same as asking $J/KJ \to K/K^2$ to be injective after localization at $\mathfrak{q}$. Equivalently, we have to show that $J_\mathfrak{q} \cap K^2_\mathfrak{q} = (KJ)_\mathfrak{q}$. By Lemma 0GH1 we know that $(K/J)_\mathfrak{q} = J'_\mathfrak{q}$ is generated by a regular sequence. Hence the desired intersection property follows from More on Algebra, Lemma 07CX (and the fact that an ideal generated by a regular sequence is H_1-regular, see More on Algebra, Section 07CU). $\square$